

Junyoung Kim

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Education

Gachon University, B.S. in Artificial Intelligence Mar. 2023 – Present

- Currently in the first semester of the fourth year
- GPA: 4.41/4.5 (Overall), 4.44/4.5 (Major)

Experience

BSURE 2025, University of Birmingham Jul. 2025 – Aug. 2025

- Developed a voice conversion model that transforms non-fluent speech into fluent speech.
- Sponsored by Google DeepMind.

Undergraduate Research Assistant, HCI Lab – Gachon University Oct. 2023 – Present

- Conducted a research project in collaboration with Seoul Asan Medical Center, focusing on web development for a medical research task.
- Developed deep learning-based techniques for recognizing user behavior patterns and interactions using video and sensor signals.
- Designed a smart wearable-based disease management model to enhance healthcare monitoring and prediction.

Publications

TriMamba: Tri-Modal Fusion and Bidirectional Mamba for Video Summarization May 2026
Junyoung Kim

- *Under Review in Conference*

Paralinguistic Acoustic Feature to Natural Language Prompt Conversion using Hybrid Transformer-LLaMA Architecture Nov 2024
Junyoung Kim, Ahyoung Choi†

- *Major Revision*

Leadership & Extracurricular Activities

DACON Competition Challenger, Dacon Mar. 2025 – Present

LG Aimers 6th, LG AI Research Jan. 2025 – Apr. 2025

Kaggle Competition Expert, Kaggle Aug. 2024 – Present

President, TakeOut Autonomous Driving Club, Gachon University Sep. 2024 – Feb. 2025

Ranker Special Lecture, DAICON Sep. 2024

Google Machine Learning Bootcamp, Google Jul. 2024 – Oct. 2024

LG Aimers 5th, LG AI Research Jul. 2024 – Sep. 2024

Vice President, TakeOut Autonomous Driving Club, Gachon University Mar. 2024 – Aug. 2024

LG Aimers 4th, LG AI Research Jan. 2024 – Apr. 2024

Member, GADIS Autonomous Driving Club, Gachon University Dec. 2023 – Present

CNN Study Mentor, Gyoha High School Oct. 2023 – Nov. 2023

Executive Member, TakeOut Autonomous Driving Club, Gachon University Sep. 2023 – Present

Honors & Awards

Data Science Competitions

Encouragement Award – 2026 Nuclear Energy AI Bootcamp Competition Feb. 2026
Korea Nuclear International Cooperation Foundation (KONICOF)

- Built an AI model for accident detection in nuclear power plants using reactor simulation data, identifying anomalies across various fault scenarios.
- Top 4

Encouragement Award – 2025 SW Centered University Digital Competition: AI Sector Aug. 2025
SW Centered University Council President Award

- Developed an AI model to classify whether a given sentence was written by a human or generated by an LLM.
- Top 4 out of 271 teams.

Encouragement Award – 2025 Bias-A-Thon Competition Jun. 2025
Director of the SKKU Intelligent Multimedia Research Center Award

- Developed a system to detect AI bias using LLaMA 3.1 8B. Since fine-tuning was not permitted by the competition rules, applied prompt engineering, few-shot learning, and LLM inference optimization.
- Top 4

Best Award – LG Aimers 6th Offline Hackathon Apr. 2025
Minister of Employment and Labor Award

- Ranked first on the leaderboard using solely deep learning models on structured data.
- Applied SAM Optimizer, F1 Loss, Weighted Brier Loss, Multi-task Learning, and Feature Embedding.
- Top 0.06% (1st out of 1,569).

Encouragement Award – Data & AI-Based Price Prediction Competition Nov. 2024
Korea Agro-Fisheries & Food Trade Corporation Director Award

- Forecasted agricultural product prices 10, 20, and 30 days ahead using a variety of features.
- Built a deep learning-based time series model leveraging approximately 30,000 data points for accurate price prediction.
- Top 0.6% (3rd out of 1,156).

Best Award – 2024 GBT Hackathon Oct. 2024

- Participated in a news classification competition using approximately 55,000 article titles and keywords.
- Compared LGBM with SVD/TF-IDF against a BERT-based model; LGBM achieved superior performance.
- Top 2.

Silver Award – 2024 AI Korean Language Proficiency Evaluation Competition Sep. 2024
National Institute of the Korean Language Director Award

- Developed a dialogue understanding model to select the most contextually appropriate response from three candidates.
- Evaluated multiple LLMs (LLaMA3-8B, Gemma2-9B, SOLAR-10.7B) and selected SOLAR-10.7B as the backbone model based on superior performance.
- Top 5.

Sponsorship Award – SW Centered University Digital Competition: AI Sector Sep. 2024
42Maru CEO Award

- Developed a model to classify speaker voices as genuine or deepfake-generated.
- Top 11.

Top 7% – Kaggle LMSYS Chatbot Arena Human Preference Predictions Aug. 2024

- Developed a model to predict which of two chatbot responses is preferred by human evaluators, using data collected from the Chatbot Arena platform.
- Used Gemma2-9B-it as the backbone model, trained on 25,000 natural language preference samples.

Top 6% – Kaggle BirdCLEF 2024

Jun. 2024

- Developed a model to predict bird species from audio recordings.
- As inference was restricted to CPU, used EfficientNet-B0 as the backbone and applied techniques to address class imbalance.

2nd Place – DACON Code Similarity Judgment AI Competition (Season 2)

Apr. 2024

- Developed a model to determine whether two C++ code snippets share the same logic.
- Evaluated multiple backbone models and selected UniXCoder based on the best performance.
- Designed the dataset to randomly sample from a large combinatorial space of code pairs.

Encouragement Award – 2023 Data Creator Camp

Oct. 2023

NIA Director Award

- Developed a model to classify 50 Korean food categories.
- Since the competition restricted the backbone to ResNet, focused on techniques beyond architectural changes to improve performance.

Grand Award – UNIST Supercomputing Camp

Jul. 2022

UNIST President Award

- Completed two tasks over four nights and five days: parallelizing code for maximum execution speed, and developing a model to detect mask-wearing.

Autonomous Driving Competitions

1st Place – 2025 Autonomous Driving AI Challenge

Nov. 2025

Minister of Science and ICT Award

- Developed a deep learning model to detect pedestrians, cars, and cyclists from 64-channel and 128-channel LiDAR data.

Encouragement Award – 2025 Student Creative Mobility Competition

Nov. 2025

Korea Transportation Safety Authority Award

- Competed in an autonomous navigation challenge using a half-scale vehicle.
- Detected and classified left and right cones from 16-channel LiDAR point clouds for lane guidance.
- Top 4 out of 23 teams.

Encouragement Award – Advanced Autonomous Driving Robot Race (1/2 Scale Division)

Jul. 2025

KEPCO KDN CEO Award

- Developed a 2D object detection model to detect ERP-42, a half-scale autonomous vehicle platform.
- Top 4.

1st Place – 2024 Autonomous Driving AI Challenge (Vehicle-Based 3D Object Detection)

Nov. 2024

Minister of Science and ICT Award

- Developed a deep learning model to detect pedestrians, cars, and cyclists from 64-channel and 128-channel LiDAR data.

1st Place – 2024 Autonomous Driving AI Challenge (Edge-Infrastructure 3D Object Detection)

Nov. 2024

IITP President Award

- Developed a deep learning model to detect pedestrians, cars, and cyclists from 32-channel LiDAR data.

Encouragement Award – 2024 Student Creative Mobility Competition

Nov. 2024

Korea Transportation Safety Authority Award

- Competed in an autonomous navigation challenge using a half-scale vehicle across various missions.
- Detected and classified left and right cones from 32-channel LiDAR point clouds for lane guidance.

Effort Award – Intermediate Autonomous Driving Mobility Race (1/5 Scale Division) Jul. 2024

- Competed in an autonomous driving race using a vehicle approximately 1/5 the size of a standard car on a designated course.

Excellence Award – HL Mando & HL Klemove Autonomous Driving Mobility Competition (VLF University Division) Oct. 2023

HL Klemove CEO Award

- Competed on a hybrid course combining line-following sections with wall-guided navigation sections.

Other Awards

Gachon Talent Award Apr. 2025

1st Place – Gachon University Extracurricular Project Competition (2024-2) Dec. 2024

1st Place – Gachon University Extracurricular Project Competition (2024-1) Jul. 2024

Scholarships

Presidential Science Scholarship Mar. 2025 – Aug. 2026

- Awarded by the President of the Republic of Korea.
- Selected as one of the top 60 recipients nationwide for developing world-class scientists in engineering.

Academic Excellence Scholarship Sep. 2023 – Feb. 2025

- Gachon University

Technologies

Languages: C++, C, SQL, Python

Technologies: PyTorch, ROS

Tools & Platforms: Git, GitHub, Docker, Linux (Ubuntu)